Umuzi Analytics

A plain-English guide: what the project means, what it found, and how to run and explore it

NISR Big Data Hackathon 2026, Track 1: Agricultural Productivity. Prepared 29 October 2026.

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1. Words to know first

The project uses some farming and statistics words. Each one is explained below with an example from the actual data. If you read only this page and Part 5, you will understand the findings.

1.1 Farming words

Word	What it means	Why it matters here
Yield	How much crop one piece of land produces. Written as **tonnes per hectare (t/ha)** or kilograms per hectare (kg/ha). 1 tonne = 1,000 kg.	It is the main number we compare. Maize in Rwanda gave about 1.8 t/ha in 2025/26: on average, 1,800 kg of maize from each hectare.
Hectare (ha)	A unit of land: a square of 100 m by 100 m (about one and a half football pitches).	Used for land area. Small areas give less reliable numbers (see 'enough evidence').
Harvested area	The land where the crop was actually harvested in that season.	Tells us how much farming a district number is based on.
Production	The total amount harvested. Production = yield x harvested area.	Used to check that NISR's numbers add up.
Productivity	Here it simply means yield: output per hectare.	The government's goal is to raise it.
Season A / Season B	Rwanda has two main farming seasons. Season A is planted around September; Season B around February to March.	They behave differently, so we only compare Season A with Season A, and B with B.
Improved seed	Certified seed of better varieties, instead of seed kept from the last harvest.	One of the two 'inputs' we look at.
Inorganic fertilizer	Factory-made fertilizer (such as NPK or DAP), as opposed to manure.	The other input.
Inputs / **input use**	Things a farmer adds to grow more: seed, fertilizer. 'Input use' here means the **percentage of farmers in a district who used it**, for all crops together.	Helps separate 'farmers lack seed and fertilizer' from 'something else is wrong'.
District	Rwanda has 30 districts, grouped into 5 provinces (Kigali City, Northern, Southern, Eastern, Western).	The level at which the tool works.

1.2 Words about the comparison we make

Word	What it means	Example
National yield	The yield for the whole country in the same season, as published by NISR.	National maize yield in Season A 2026: 1.99 t/ha.
Yield ratio	District yield divided by national yield. 1.00 = the same as the country. 0.69 = 69% of it.	Ruhango maize 2026A: 1.38 / 1.99 = 0.69.
Below national	In this project: the yield ratio is under 0.90, meaning the district is more than 10% below the national yield. Small differences are ignored because survey numbers are never exact.	0.69 is below national; 0.95 is not.
Shortfall	Simply: falling short. A district has a shortfall when its yield is below national (as defined above). A national shortfall means the country is below the government target.	Maize is 13% short of its 2025/26 target.
Persistent shortfall	Below national in **every** season we have (at least 2) of the same type. The problem keeps coming back.	Ruhango maize: below in 3 of 3 Season A releases.
Latest-season dip	Below national in the newest season only. Could be one bad season (rain, pests).	3 districts for maize, Season A.
Enough evidence (adequate coverage)	The district harvested at least 500 ha of the crop in that season. Below that, the survey measured few fields, so the number is shaky.	Paddy rice: most districts grow little, so many are 'not enough evidence'.
Target	The yield the government plans to reach each year, written in the PSTA 5 plan.	Maize target for 2025/26: 2.06 t/ha.

1.3 Organisations and documents

Name	What it is
NISR	National Institute of Statistics of Rwanda: the government office that collects and publishes official statistics.
SAS	Seasonal Agricultural Survey: NISR's survey of farms every season. It publishes yields, areas and input use per district.
Release	One SAS publication for one season, for example 'SAS-2026A' = Season A of 2026.
MINAGRI	Ministry of Agriculture and Animal Resources.
PSTA 5	MINAGRI's 5-year agriculture plan (2024/25 to 2028/29). Page 21 lists yield targets for 8 priority crops.
NST2	Rwanda's National Strategy for Transformation 2 (2024 to 2029). It aims to raise crop productivity by 50%.

1.4 Statistics words (you only need the idea)

Word	The idea in one sentence
Correlation (we use the 'rank' kind, called Spearman)	A number from -1 to 1 saying how closely two lists move together. Here: if we order the districts from best to worst in one season, are they in nearly the same order the next season? 1 = exactly the same order, 0 = no relation.
Chance / random	We ask: if the districts were shuffled like cards, how often would we see a pattern this strong? If almost never, the pattern is real, not luck.
p-value	The answer to that question as a fraction. p = 0.0005 means 'about 5 times in 10,000 by luck'. Small p (below 0.05) = unlikely to be luck.
Correction for many tests (Holm)	When you run many tests, a few look good by luck. This correction makes the bar higher so we do not fool ourselves.
Association vs cause	Two things moving together (association) does not prove one causes the other. All our input findings are associations.
Threshold	A cut-off line we chose, such as 90% of national or 500 ha. You can move them in the app.
Sensitivity check	Re-running the analysis with different thresholds to see if the answer changes.
Machine learning (ML)	Software that learns to predict from many examples. We did **not** use it (Part 4.5 explains why).

1.5 Software words

Word	What it means
Pipeline	The chain of steps that turns a downloaded file into what you see on screen.
Validation check (in the code: a 'gate')	An automatic test of a new data file. If it fails, the file is not used.
Snapshot	A saved, numbered copy of all the data at one moment. The app always reads exactly one snapshot.
Active snapshot	The snapshot the app is currently showing. Recorded in the file `data/snapshots/ACTIVE.json`.
Quarantine	A folder where rejected files are put aside with the reason, instead of being used.
Rollback	Going back to an earlier snapshot if a mistake is found. Nothing is deleted.
Streamlit	The free Python tool used to build the app. It runs a small web page on your own computer.
Python / virtual environment	Python is the programming language. A virtual environment is a private folder holding the exact add-on packages the project needs.
Automated tests	Small programs that check the code still gives correct answers. One command runs all 38.

2. The data we used

Every number in the app comes from a real, downloaded, public file. Nothing was made up. When the project started there was no data in the folder at all, so all of it was found on official websites.

Data	Who publishes it	What we took from it	Where it is in the folder
SAS tables, 5 seasons: 2024 A, 2024 B, 2025 A, 2026 A, 2026 B (one Excel file each, about 35 tables inside)	NISR	For each district and crop: yield, harvested area, planted area, production. For each district: % of farmers using improved seed, factory fertilizer, manure, erosion protection	`data/raw/nisr_sas/`
SAS 2024 yearly workbook	NISR	Only to double-check the 2024 files (they matched exactly)	`data/raw/reference/`
SAS 2026A report (PDF)	NISR	How the survey works: about 1,200 sample areas; yields are estimated by the farmers	Cited only
PSTA 5 plan (PDF), page 21	MINAGRI	National yield targets for 8 priority crops, each year to 2028/29	`data/reference/psta5_productivity_targets.csv`
District boundaries	geoBoundaries (made from NISR maps)	The shapes of the 30 districts, to draw the map	`data/raw/boundaries/`

Licence. NISR data is published under CC BY 4.0: anyone may use and share it if they credit NISR. The app credits 'National Institute of Statistics of Rwanda, Seasonal Agricultural Survey' on every chart.

2.1 What is missing

- **2025 Season B** could not be found on the NISR website. The app shows it as missing; it is not filled in.
- Seasons before 2024 had no downloadable tables on the NISR pages we checked.
- No rainfall, soil, price or market data was added. So the tool cannot say *why* a yield is low; it can only point to where to look.

2.2 Five things about this data you should know

1. **NISR changes table numbers between releases.** The yield table is Table 10 in one file and Table 13 in another. So the software finds tables by their title, not their number.
2. **A printed 0 usually means 'not measured', not 'zero harvest'.** NISR prints 0 when a crop was not seen in a district's sample. Treating it as a real zero would invent disasters, so these are stored as 'not reported'.
3. **The national total row has different names** ('National', 'Total', 'S/Total'). An early version of our code missed two of these names, which made a wrong conclusion look right (Part 4.4 tells the story).
4. **Yields are farmers' own estimates**, and NISR only publishes how precise the numbers are for the country, not for each district. That is why we ignore differences under 10% and ask for at least 500 ha.
5. **Input use is for all crops together**, not for maize or beans separately.

2.3 Checks we ran on the data

Check	Result
Does production = yield x harvested area?	Yes, in all 2,587 cells where all three numbers are present
Do the separate 2024 files match the 2024 yearly workbook?	Yes, exactly
How many cells were printed as 0?	125 to 225 per release; all stored as 'not reported'
How many district numbers rest on less than 500 ha?	About a third (34%); these are flagged, not hidden

3. The aim of the project

3.1 The problem in plain words

The government wants farms to produce much more per hectare (NST2 aims for a 50% increase in productivity). The PSTA 5 plan sets a yield target for each of 8 priority crops. In 2025/26, **all 8 crops were below their target** at national level.

But the targets are national, and the work is done in districts. A district agronomist (the government agriculture officer of a district) planning the next season faces three problems:

- The SAS gives one Excel file per season with hundreds of numbers. Comparing seasons by hand is slow.
- From one season alone, you cannot tell **a district that is always behind** from **a district that had one bad season**. These need different responses.
- Even when a district is always behind, it is not clear whether farmers lack seed and fertilizer, or whether the problem is something else (soil, erosion, variety, water). Sending more fertilizer to the wrong place wastes money.

3.2 The aim

Help district and national planners decide where to look first, and what to check first, for each crop. In one sentence, the question the app answers is:

> *"Where are district crop yields persistently below the national yield, and what should be checked first?"*

3.3 Who it is for

User	What they get
District agronomist	Which crops in their district are always behind, compared with the country, and a suggested first check
MINAGRI or RAB planner	A map and a grid showing all districts and crops at once, to choose where to send field teams
Anyone checking the numbers	Every number traced to the exact NISR file and table

3.4 What the tool does NOT do

- It does not predict next season's yield.
- It does not say 'give this district more fertilizer'. It gives a **route for investigation**, which a person on the ground must confirm.
- It does not prove causes. It shows where things happen together.
- It does not judge whether a crop suits a district. 'Below national' is not the same as 'below what that land could do'.

4. How the system works

4.1 The journey of the data

Think of it like a kitchen: ingredients are delivered (download), unpacked (read), inspected (check), stored with a date label (snapshot), and only served when the chef approves (approve). The app only ever shows food that passed inspection.

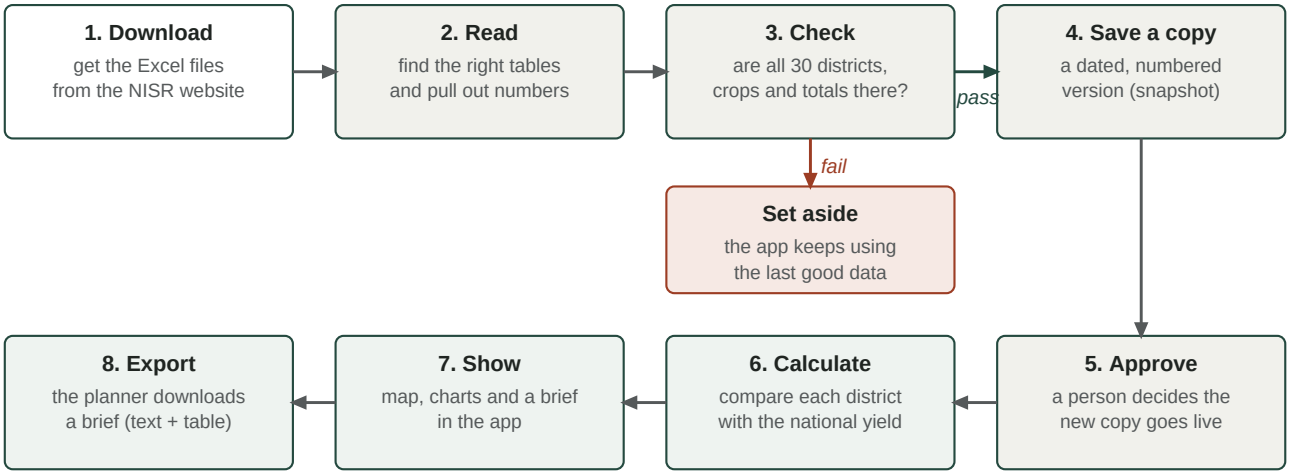


Figure 1. The eight steps. A file that fails the checks is set aside and never reaches the app.

Step	In plain words	Program that does it
1. Download	Fetches the NISR Excel files. A 'fingerprint' (checksum) of each file is stored, so we notice if NISR ever changes a file.	`scripts/download_sources.py`
2. Read	Opens each Excel file, finds the yield, area, production and input tables by their titles, and turns them into one long, tidy list of numbers.	`src/ingestion/sas_adapter.py`
3. Check	Are all 30 districts there? All needed crops? The national row? Does production = yield x area? Any duplicates? If anything important fails, the file is set aside with a note.	`src/validation/gates.py`
4. Save a copy	Saves a new numbered snapshot with all the data, plus a record of where it came from.	`src/services/snapshot_store.py`
5. Approve	A person runs one command to make the new snapshot live. Mistakes can be undone (rollback).	`scripts/refresh_data.py`
6. Calculate	Works out each district's yield ratio, checks every season, and gives each district a label (4.2).	`src/analytics/indicators.py`
7. Show	Draws the map and charts, and builds the app's eight screens.	`src/viz/charts.py, app.py`
8. Export	Creates the Evidence Brief: a short text and table a planner can share.	`src/decisions/brief.py`

Why this design? When NISR publishes a new season, nobody needs to change the app. Someone downloads the file and runs two commands; the checks make sure a bad file cannot break anything. This was tested with a made-up file.

4.2 How each district gets its label

For the crop and season you choose, every district gets one of five labels. The computer answers simple yes/no questions in this order:

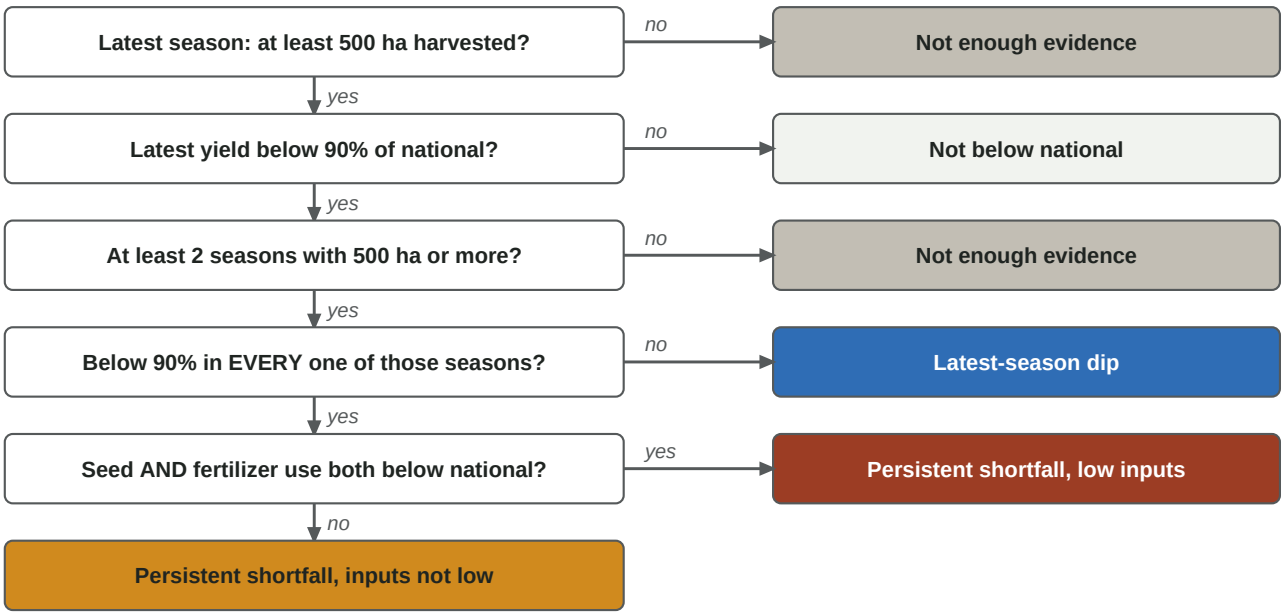


Figure 2. The labelling rules. If seed or fertilizer data is missing, the district is also marked 'Not enough evidence': a missing number is never treated as good news.

Label (colour on the map)	What it means	Suggested first check
Persistent shortfall, low inputs (dark red)	Always behind, and both improved seed and fertilizer are used by fewer farmers than nationally	Can farmers get and afford seed and fertilizer on time?
Persistent shortfall, inputs not low (orange)	Always behind, even though seed or fertilizer use (or both) is at or above the national level	Something else: soil quality and acidity, erosion, seed variety, water
Latest-season dip (blue)	Behind only in the newest season	Was there a bad season (rain, pests, late planting)? Watch the next release
Not enough evidence (grey)	Too little land or too few seasons to judge	Check on the ground before acting
Not below national (white)	No shortfall	Could be a good example for neighbours

4.3 A worked example: Ruhango, maize, Season A

Question	Ruhango's numbers	Answer
Enough land in the latest season?	4,026 ha harvested (need 500)	Yes
Below 90% of national now?	1.38 t/ha vs national 1.99 t/ha = ratio 0.69	Yes
At least 2 seasons with enough land?	3 Season A releases (2024, 2025, 2026)	Yes
Below in every one?	Below 90% in 3 of 3	Yes: persistent
Seed and fertilizer both below national?	Improved seed 29.3% of farmers (national 37.0%); fertilizer 39.3% (national 67.2%)	Yes
Label		**Persistent shortfall, low inputs**: check seed and fertilizer access first

4.4 A mistake we found, and why the checks matter

In an early version, the code did not recognise the national row when NISR named it 'Total' or 'S/Total'. As a result, no district ever looked like it had low input use, and the app said every struggling district had 'inputs not low'. That looked believable but was wrong. It was caught because the result seemed too neat. The code was fixed, and a new check now rejects any file where the national row is missing, so the same mistake cannot happen silently again.

4.5 How we checked that the patterns are real

Before looking at any results, we wrote down three questions (hypotheses) and how we would test them. Writing them first stops anyone from picking only the tests that happen to look good. The questions were:

#	Question	How it was tested	Outcome
H1	Do districts keep the same position (ahead or behind) from season to season?	Rank correlation between seasons, then 5,000 random shuffles	**Yes**, strongly
H2	Are 'always behind' districts more common than luck would give?	Count them, and compare with 2,000 shuffled versions	**Yes**: 60 vs about 17
H3	Do districts with more seed and fertilizer use have better maize and bean yields?	Rank correlation, corrected for many tests	**Beans yes; maize no**

Rainfall, soil and prices could not be tested because that data was not part of the project.

4.6 Why there is no machine-learning or AI model

- Each district has only 3 Season A and 2 Season B numbers per crop. That is far too few to train and fairly test a model.
- A very simple rule, 'a district stays where it was last season', is already almost always right (H1). A model could not prove it does better.
- The planner's decision is 'where to look and for what', not 'what exact yield next season'.
- Simple rules can be explained to anyone and checked by hand, as the Ruhango example shows.

In short: we chose clear rules plus honest statistical checks over a complex model the data cannot support.

5. Results and insights

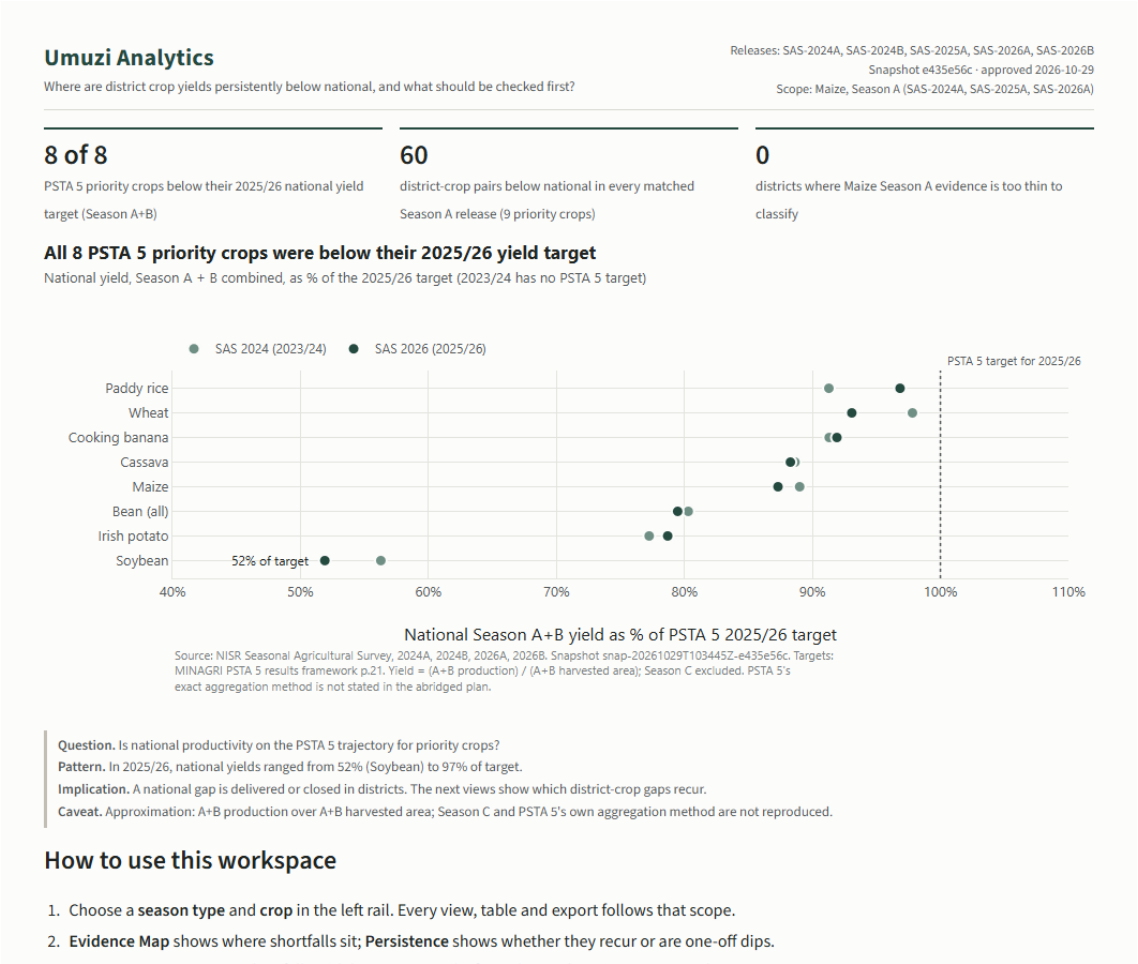
Each finding below says what we found, what it means, and what a planner could do with it.

Finding 1: Every priority crop is behind its national target

Crop	National yield 2025/26 (t/ha)	Target (t/ha)	Reached
Soybean	0.47	0.9	52%
Bean	0.72	0.9	79%
Irish potato	7.95	10.1	79%
Maize	1.80	2.06	87%
Cassava	13.86	15.7	88%
Cooking banana	15.72	17.1	92%
Wheat	1.21	1.3	93%
Paddy rice	4.36	4.5	97%

Season A and B combined. Season A maize yield did not grow at all: 2.04, 1.99 and 1.99 t/ha in 2024, 2025 and 2026.

What it means: the country is not on track, so knowing where to act first is useful now. **For a planner:** soybean and beans are furthest from target.



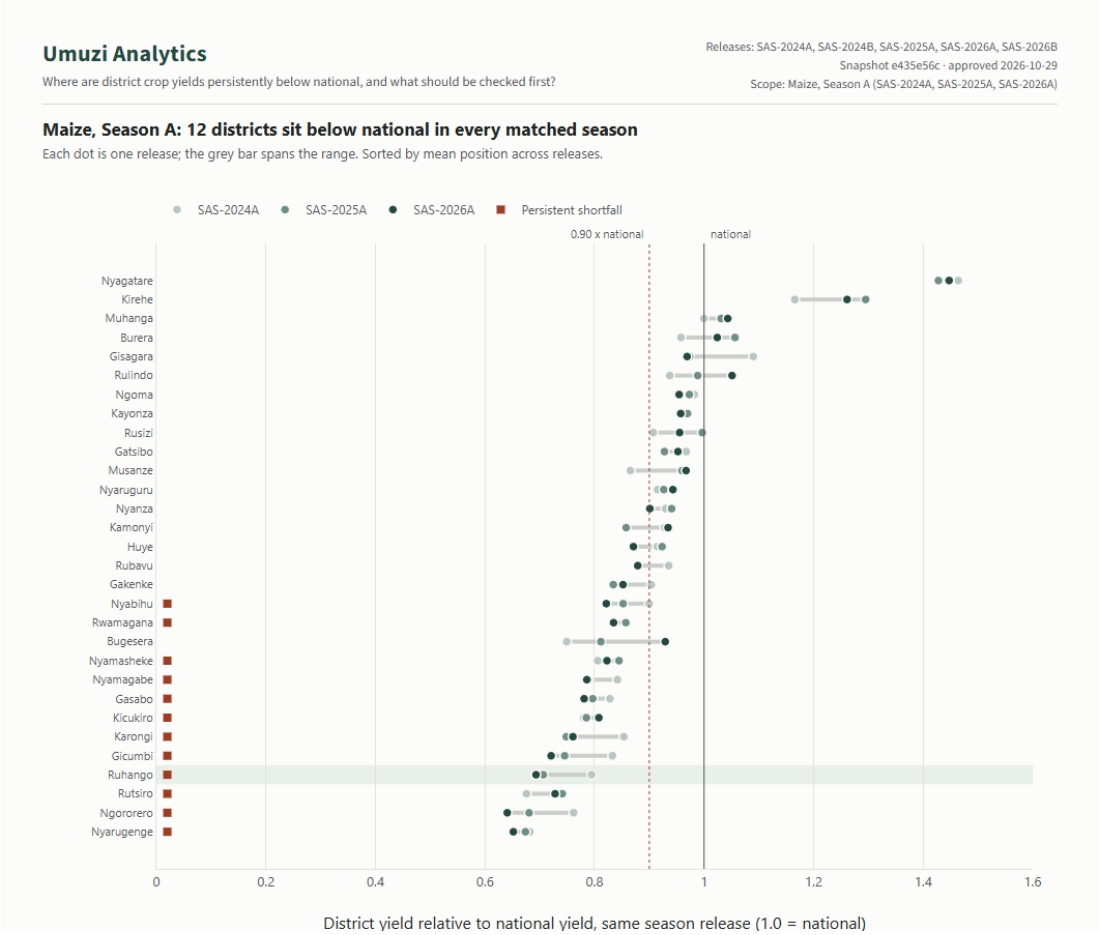
The Overview screen shows this as a dot chart: each dot is a crop, the further left, the further from target.

Finding 2: The same districts stay behind, season after season

- If you rank districts from best to worst yield, the order barely changes between seasons (correlations of 0.70 to 0.99 for 7 crops; much stronger than chance).
- 60 district-and-crop combinations were below national in all three Season A releases. If seasons were random, we would expect about 17. The chance of seeing 60 by luck is about 5 in 10,000.
- For maize in Season A, 12 of 30 districts are persistently behind; 3 more had just a latest-season dip.

What it means: low yields are mostly a lasting, local problem, not bad luck in one season. **For a planner:** persistently behind districts are the best places to send a field diagnosis team, because the problem is likely to still be there next season.

Caution: NISR does not say whether it visits the same sample areas every season. If it does, part of this stability comes from the survey design.

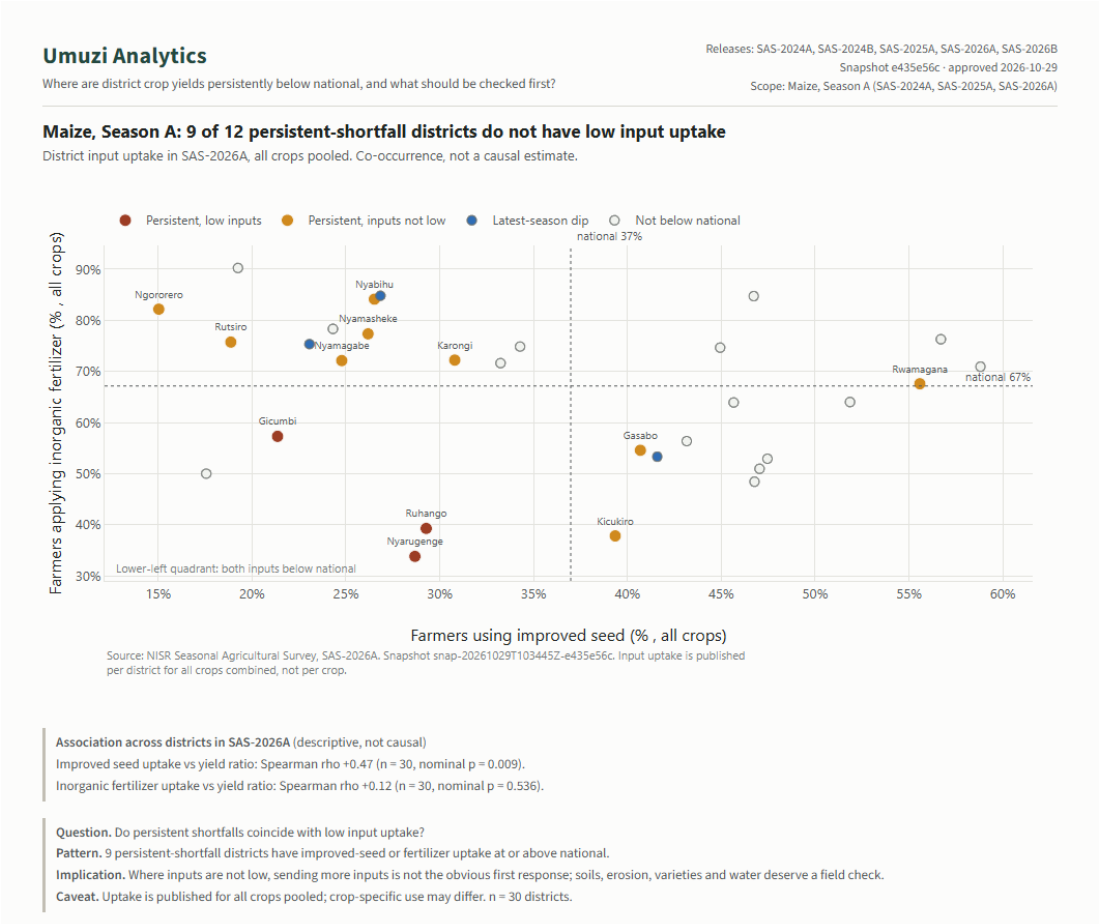


The Persistence screen. Each row is a district, each dot one season. Dots left of the dotted line (0.90) are below national. Dark squares mark districts behind in every season.

Finding 3 (the most useful): most persistently low maize districts do NOT lack seed and fertilizer

- Of the 12 maize districts always behind in Season A, only **3** (Gicumbi, Ruhango, Nyarugenge) have both low seed use and low fertilizer use.
- The other **9** do not have *both* inputs low: each has at least one (seed or fertilizer) at or above the national level. Six of them (five districts in Western Province plus Nyamagabe) are a striking group: **more** farmers there use fertilizer than the national average (72% to 84%, vs 67%), but **fewer** use improved seed (15% to 31%, vs 37%).
- For beans, districts where more farmers use fertilizer clearly have higher yields, in all 5 seasons. For maize, the link with seed or fertilizer use is weak and could be luck.

What it means: the common idea 'low yield = farmers need more fertilizer' does not fit most of these districts. **For a planner:** in those 9 districts, the first questions should be about seed varieties, soil (for example acidity), slopes and erosion, not only about delivering more fertilizer. In Gicumbi, Ruhango and Nyarugenge, input access is the first thing to check.



The Input Context screen. Across: % of farmers using improved seed; up: % using fertilizer. The dotted lines are the national values. The bottom-left box = both below national.

Finding 4: Other useful patterns

- **Season A and Season B are different stories.** A district's maize position in Season A says little about Season B (correlation 0.24). Never mix the two.
- **Some districts are behind in many crops at once:** 10 districts in Season A are persistently behind in 3 or more priority crops (for example Ruhango, Kayonza, Rwamagana, Nyamagabe). A district-wide issue is

more likely there.

- **Careful with Irish potato:** many 'behind' districts are in the warm, low-lying east, where potato is not a main crop. Behind the national figure does not always mean underperforming.
- **The 90% line matters:** moving it to 80% or 95% changes the label of about a quarter to a third of maize districts. The 500 ha rule matters much less. The app lets you test this yourself.

6. Do the insights make sense, and can they make an impact?

A finding is only useful if it is true, new and leads to a decision. Here is an honest assessment of each.

Insight	Is it solid?	Is it new?	Can it change a decision?
All 8 crops behind target	**Solid.** Straight from official NISR yields and the official target table.	Partly. Officials may know it, but it is rarely shown side by side.	Sets the urgency; does not by itself say where to act.
Same districts stay behind	**Solid.** Very strong statistically, and in 7 crops. One caution about the survey design (Part 5).	Yes. The SAS reports publish one season at a time and never show this.	**Yes.** Tells planners where a field visit is likely to find a lasting problem.
Most persistent maize districts are not input-poor	**Medium.** The numbers are correct, but input use covers all crops, not maize alone, and it is association, not cause.	Yes. It contradicts the easy assumption.	**Yes, potentially the biggest.** It could stop money going to the wrong fix, and send soil and seed experts instead.
Beans respond to fertilizer use	**Fairly solid.** Strong in all 5 seasons, but still association.	Expected by agronomists; useful confirmation.	Supports fertilizer programmes for beans.
Season A and B differ	**Solid.**	A method lesson more than a discovery.	Prevents wrong averages.

6.1 Why the insights make sense

- Every number can be traced back to a cell in an official NISR file, and tests check some of these by hand (for example, Nyagatare maize 2026A = 2,886.67 kg/ha).
- The questions were written down before the analysis, and the tests were corrected for luck.
- The results are consistent across seasons and crops, not a one-off.
- The limits are stated next to the results: farmer-estimated yields, no district error margins, all-crop input data.

6.2 How it could make an impact

1. **Better targeting of field visits.** Instead of reading 30 x 27 numbers per season, an agronomist sees in one screen which crops are persistently behind in their district.
2. **Better choice of first check.** Inputs, or soil and variety? The two routes lead to different experts and budgets.
3. **Faster updates.** When NISR publishes a new season, the app updates without any programming.
4. **Shareable evidence.** The Evidence Brief puts the numbers, the source and the questions to ask on one page.

6.3 What is still unproven

The tool has **not yet been tested with real planners**, and no field visit has yet confirmed a label. So we cannot claim it saves time or raises yields. `docs/field_validation_plan.md` describes how to test this: pick a few districts per label, visit them, and record whether the suggested first check was the right one. Say this openly in the presentation: it is honest, and judges value it.

7. Start the app with one command

A ready-to-use Python environment already exists on this computer at `C:\Users\jeanc\.venvs\umuzi`. (It is kept outside the Box folder so Box does not try to sync thousands of files.)

7.1 The command

Open **PowerShell** (or the terminal in VS Code), paste this single line, and press Enter:

```
cd C:\Users\jeanc\Box\Big-Projects\NISR; C:\Users\jeanc\.venvs\umuzi\Scripts\python -m streamlit run app.py
```

7.2 What you will see

1. The terminal prints *'You can now view your Streamlit app in your browser'* and a link `http://localhost:8501`.
2. Your browser opens that page by itself (if not, click the link). The first load takes a few seconds.
3. The app opens on the **Overview** screen.
4. To stop the app: click in the terminal and press **Ctrl + C**.

'localhost' means the app runs only on your own computer. Nobody else can see it until it is deployed online, which has not been done.

7.3 If it does not start

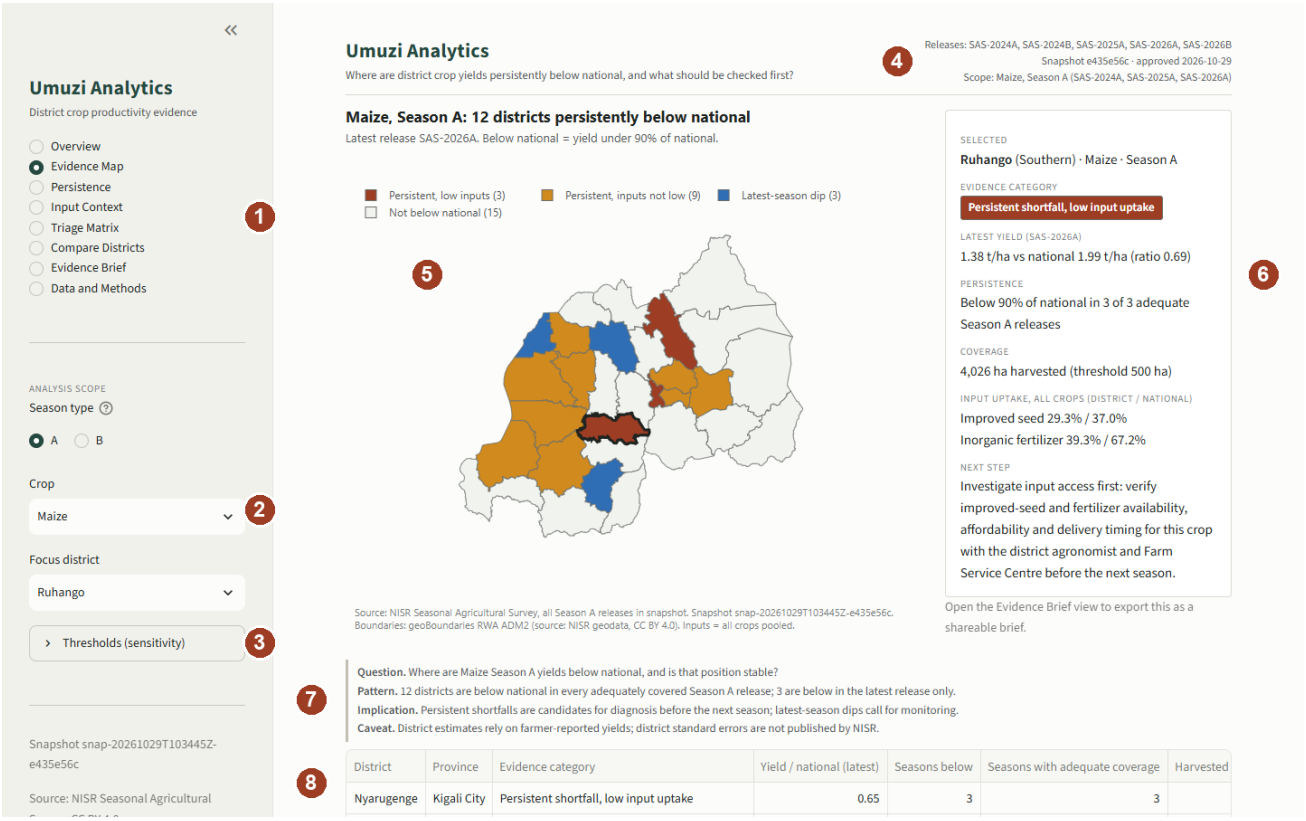
Message	What to do
'...python' is not recognized / cannot find path	The environment is missing. Create a new one (below).
Port 8501 is already in use	The app is already running in another window. Use that, or close it with Ctrl + C.
'No approved data snapshot is active'	The data/snapshots folder is incomplete or ACTIVE.json is missing. Run the status command in Part 8.4, then approve the snapshot it lists (README.md).

To create a fresh environment on any computer (Python 3.11 or newer must be installed), run these once:

```
cd C:\Users\jeanc\Box\Big-Projects\NISR
python -m venv C:\Users\jeanc\.venvs\umuzi2
C:\Users\jeanc\.venvs\umuzi2\Scripts\python -m pip install -r requirements-dev.txt
```

Then use the one-line command above with `umuzi2` instead of `umuzi`.

8. Find your way in the app



The app window (Evidence Map screen). The numbers are explained below.

#	Part of the screen	What it does
1	**Screen list** (left)	Click to switch between the 8 screens.
2	**Your choices**: season, crop, focus district	Everything on every screen follows these. 'Focus district' is the district you want to study closely.
3	**Thresholds**	Click to open. Move the 90% line or the 500 ha rule to see how labels change. A notice then warns that you are not using the standard settings.
4	**Data label** (top right)	Which NISR seasons are loaded and which snapshot you are looking at.
5	**Map**	Each district coloured by its label. Your focus district has a thick black outline. Hover over a district to see its numbers.
6	**Evidence panel**	The full story for your focus district: label, yields, seasons behind, land, inputs, and the suggested first check.
7	**Reading box**	Under each chart: the question, the pattern, what it implies, and a caution.
8	**Table**	All districts with their numbers. Click a column header to sort. There is a CSV download button.

8.1 Which screen answers which question

If you want to know...	Go to	Look at
How far is Rwanda from its yield targets?	**Overview**	The dot chart: crops on the left are furthest from target
Where are the problem districts for a crop?	**Evidence Map**	Dark red and orange districts
Is a district always behind, or just once?	**Persistence**	Its row: are all dots left of the dotted line?
Is the problem seed and fertilizer, or something else?	**Input Context**	Bottom-left box = both inputs low
Is a district behind in many crops at once?	**Triage Matrix**	Rows with many coloured cells
How does my district compare with another?	**Compare Districts**	Two lines against the national line; read any warning about thin data
What do I send to my manager or field team?	**Evidence Brief**	The one-page brief; download as text or table
Can I trust these numbers? Where do they come from?	**Data and Methods**	Data status, checks, definitions, sources

8.2 A 10-minute demo route (also in `demo/demo_script.md`)

Keep the default choices (Season A, Maize, Ruhango) and follow these steps. The right-hand column is what you should see.

Step	Do this	You should see
1	Stay on **Overview**	'8 of 8' crops below target; soybean furthest left at 52%
2	Click **Evidence Map**	Title '12 districts persistently below national'; Ruhango outlined in black; panel says 'Persistent shortfall, low input uptake', 1.38 vs 1.99 t/ha
3	Click **Persistence**	Ruhango's row highlighted, all 3 dots left of the line
4	Click **Input Context**	Title '9 of 12 ... do not have low input uptake'
5	Click **Triage Matrix**	'10 districts show persistent shortfalls in three or more priority crops'
6	Click **Compare Districts**, pick Nyarugenge as the second district	Both lines under the national line in every season
7	Click **Evidence Brief**, then 'Download brief (Markdown)'	A file named umuzi_brief_Ruhango_maize_A.md is downloaded
8	Click **Data and Methods**	A yellow note that 2025 Season B is missing; 5 releases passed the checks

Try also: switch Season to B (the numbers change); choose Crop = Bean; or open Thresholds and move the 90% line to 80% and watch districts change colour.

8.3 What the app labels mean on screen

The app uses the official wording: 'Persistent shortfall, low input uptake' (= low inputs), 'Persistent shortfall, inputs at or above national' (= inputs not low), 'Latest-season dip', 'Insufficient evidence - field verification required' (= not enough evidence), and 'Not below national'. 'Input uptake' simply means input use.

8.4 Optional commands for adding new data

These run in the same terminal, from the project folder. Replace *python* with `C:\Users\jeanc\.venvs\umuzi\Scripts\python`.

Command	What it does	What you should see
`python -m scripts.refresh_data status`	Shows which data is live	Active snapshot snap-20261029T103445Z-e435e56c, 5 releases, 10 snapshots
`python -m scripts.refresh_data ingest ...`	Adds a new NISR season file (full example in README.md)	A new snapshot id, or 'quarantined' with the reason
`python -m scripts.refresh_data approve --snapshot <id>`	Makes that snapshot live	The app shows the new season
`python -m scripts.run_analysis`	Re-runs all the statistics	Results printed; files in reports/ updated (25 seconds to a few minutes)

9. The project folders

Everything is in `C:\Users\jeanc\Box\Big-Projects\NISR`. The tree below shows each folder in plain words.

```
NISR\
|-- app.py ..................... the app itself (the 8 screens)
|-- README.md .................. short project description and commands
|-- guide.pdf .................. this guide
|-- EXECUTION_LOG.md ........... diary of what was done and what went wrong
|-- requirements.txt ........... list of Python packages the app needs
|-- requirements-dev.txt ....... the same, plus packages for the tests
|
|-- src\ ....................... the engine (the code the app uses)
|   |-- config.py .............. settings: districts, crop names, 90% and 500 ha
|   |-- ingestion\ ............. step 2: reading NISR Excel files
|   |-- validation\ ............ step 3: the checks
|   |-- services\ .............. steps 4-5: snapshots, approval, data status
|   |-- analytics\ ............. step 6: ALL formulas and labelling rules
|   |-- viz\ ................... step 7: every chart and the map
|   |-- decisions\ ............. step 8: the Evidence Brief
|
|-- scripts\ ................... commands you run by hand
|   |-- download_sources.py .... download NISR files again
|   |-- refresh_data.py ........ add, approve or undo data
|   |-- run_analysis.py ........ repeat the statistics, save results
|
|-- data\
|   |-- raw\ ................... original downloaded files (never edited)
|   |-- reference\ ............. PSTA 5 targets, typed in from page 21
|   |-- snapshots\ ............. saved data versions; ACTIVE.json = the live one
|   |-- incoming\, quarantine\, staging\ ... working folders (not published)
|   |-- sources.json ........... the web addresses of every file
|   |-- release_registry.json .. list of every file loaded, with fingerprint
|
|-- tests\ ..................... the 38 automatic tests
|-- docs\ ...................... written explanations (16 documents)
|-- reports\ ................... findings, statistics results, chart files
|-- demo\ ...................... script for the 3-minute demo video
|-- submission_private\ ........ for student cards; never published
```

9.1 Which file to open for what

I want to...	Open
Understand the project in 5 minutes	`README.md`
See the findings with all numbers	`reports/eda_findings.md`
Read every limitation	`docs/limitations.md`
Understand every column of the data	`docs/data_dictionary.md`
See why this problem was chosen over others	`docs/problem_discovery.md`
Read the method in detail	`docs/methodology.md`, `reports/analytical_methods.md`
Know what the competition requires and whether we meet it	`docs/compliance_matrix.md`, `reports/submission_checklist.md`
Prepare the demo video	`demo/demo_script.md`
Put the app online (when you decide to)	`docs/deployment.md`
Add a new NISR season	`docs/operations_runbook.md`
Change the 90% or 500 ha defaults	`src/config.py`
Change how labels are decided	`src/analytics/indicators.py` (function `evidence_table`)
Change how a screen looks	`app.py` and `src/viz/charts.py`
Check a number against NISR	The original Excel file in `data/raw/nisr_sas/`

Good to know: `data/snapshots/` holds 10 snapshots. Five were made by the early, faulty code version (Part 4.4). They are kept on purpose as history and are not used. You may delete them by hand if you prefer.

10. The tests: run them and know what to expect

The tests are small programs that check the system still gives the right answers. Run them before a demo or after any change. One command, from the project folder:

```
cd C:\Users\jeanc\Box\Big-Projects\NISR; C:\Users\jeanc\.venvs\umuzi\Scripts\python -m pytest
```

What a good result looks like: a row of 38 dots (one per test), then a line similar to `38 passed in ...s`, where ... is the time taken (normally 20 to 40 seconds). No internet is needed, and no browser opens: the app tests run it invisibly. Warnings printed by libraries are not failures; only lines that say **failed** or **error** matter.

Test file	Tests	What it checks, in plain words
`tests/test_pipeline.py`	8	The data steps work: tables are found even when renumbered; 'Total' rows are read; a printed 0 is not treated as zero; a new season appears without changing the app; loading the same file twice does nothing; bad files are set aside; corrections and rollback work. *Uses made-up practice files that are clearly labelled as such and never shown in the app.*
`tests/test_analytics.py`	4	Each label is given correctly; missing input data never counts as 'not low'; moving thresholds changes results; the ratio formula is right.
`tests/test_real_data.py`	8	The real data is complete (5 seasons, 30 districts); files have not changed; production = yield x area; known numbers match NISR's files; the brief matches the table.
`tests/test_app.py`	18	All 8 screens open without error in Season A and B (16 tests); your choices carry through to the header, brief and downloads; the non-standard threshold notice appears.

10.1 If a test fails

- Read the line under 'FAILED': it names the test and shows what was expected and what came out.
- 'no active snapshot' or tests marked skipped: run `python -m scripts.refresh_data status` (Part 8.4).
- An app test timed out on a slow or busy computer: run the command again.
- Failures after you added a new NISR season: the data checks usually explain the problem; see `docs/operations_runbook.md`.

11. What is left to do (only your team can do these)

Nothing has been put online, pushed to GitHub or submitted. These steps need your accounts and your decision. Deadline: **30 October 2026, 11:59 PM** (the competition document does not state a time zone).

#	Action	Where to read more
1	Both team members read the rule that all rights in the work go to NISR	Competition PDF, page 3
2	Fill in the TEAM parts of the AI-use statement	`docs/AI_USE_DISCLOSURE.md`
3	Confirm you are eligible (2 students, at least one Rwandan, registered by 20 October)	`docs/compliance_matrix.md`
4	Decide how to share student cards (ask the organisers before making them public)	`submission_private/README.md`
5	Create a public GitHub repository and upload the project	`docs/deployment.md`
6	Put the app online (Streamlit Community Cloud) and test it	`docs/deployment.md`
7	Record the demo video (3 minutes maximum, screen recording)	`demo/demo_script.md`
8	Submit the form; open the app shortly before the deadline so it is awake	`reports/submission_checklist.md`

Known small issue: on a narrow window (about 900 pixels wide) with the left panel open, the map may not draw. Widen the window or close the left panel.